A Novel Seizure Prediction method based on Generative features

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Abstract. The diagnosis of epilepsy in hospital is mostly judged by experienced medical personnel visually observing brain waves combined with some characteristic clinical manifestations.As brain signal's complexity the understanding of EEG signal still remains challenge. In this paper we proposed a novel seizure prediction method based on generative features. Then predict seizures according to the EEG information of the epileptic patients. And we use the Extreme learning machine as the classifier of generative features. Finally, we get the highest accuracy score of 98%.

Keywords: Epilepsy; Seizure Prediction; deep learning; Neural Network

1 Introduction

Research of brain science is one of the most challenging frontier science problem. In recent years, brain research programs are constantly launched and pushed forward around the world. Epilepsy is a chronic neurologic disease of the brain caused by sudden abnormal discharge of the brain neurons. As a common disease, there are about 0.6%-0.8% population in the world suffer from epilepsy [1]. There are about six to nine million epileptic patients in china, and they are growing at the rate about four hundred thousand per year[2]. The epileptic seizures' onset time is uncertain. When it occurs, if the patient can't get appropriate treatment in time, it may be very dangerous for patients. Even a short seizure may cause severe physiological and psychological damage. If we can predict epileptic seizures automatically and in time, it will help patients to avoid many times damage caused by seizures.The research of epileptic seizures is firstly carried out in 1970s[3]. In recent years, the method of predicting epilepsy based on artificial intelligence attracts a lot of attention.

EEG-based epilepsy prediction mainly includes two steps: feature extraction and classification. When working on feature extraction, researchers can select the common method from time-domain analysis [9-11], frequency domain analysis [12-15], time-frequency domain analysis [16-17] and non-linear analysis [18-19]

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methods etc. The time-domain analysis is to extract waveform characteristics through the analysis of time-domain waveform EEG signals, such as cycle and rhythm [20]. Wavelet Transform, and Hilbert transform [16] are widely used in time frequency analysis. Hybrid dynamics method is proposed in 1990s, which includes maximum lyapunov Index [18], density correlation and kinetic similarity index [21].

Pattern recognition and machine learning methods are frequency used with classification mission.The common used method includes Support Vector Machine(SVM), Decision Tree(DT), Bayesian, etc. Subasi proposed a general EEG processing and analysis system[10].In this work PCA, ICA and LDA is used to extract features and SVM is used as classification. In 2000, Petrosian combined the original EEG signal and the wavelet decomposition coefficient as the input to RNN [25], which was the first time for RNN be used in EEG processing jobs. In 2015, Achilles [27] used two CNN structures to determine the seizure period in video. Although these studies have achieved some results, the data of individual cases are insufficient because of the obvious difference in body size and the lack of pathological data. How to effectively use a small amount of epileptic data to fully excavate the discriminant characteristics of epileptic EEG data is the key problem to solve the epilepsy prediction.

In this paper, the generation of network model GAN is introduced into the prediction of epilepsy. We propose a new EEG analysis method based on generation features, and practice the training and testing job on it. The main work of this paper are : (1)Transfer the raw EEG data into spectrum pictures (2) Augment the feature data by DCGAN model.(3) Classify the data with ELM.The remainder of this paper is structured as follows: Section 2 describes the framework we use. And the results is shown in Sections 3. Finally, followed by acknowlegments and possible future works.

1.1 Seizure Prediction method based on Generative features

1.2 Experiment Process

EEG signals are characterized by non-stationary and non-linear. Tt is difficult to carry out the data analysis directly on the raw data. Epileptic EEG signal expresses continuity in the time domain and the frequency increases in the frequency domain when seizures onset. Therefore, both the time and frequency domain information should be considered in the feature processing. The epileptic form discharge shows the typical output signal during the seizure period which reflects the energy change. This change can be used as a significant feature of epilepsy detection and prediction. In this paper, we use the short-time Fourier transform method to deal with the original EEG data result in spectrum patterns. Then take advantage of the DCGAN's function of generate new data with the real data.Finally, the ELM is used to classify the combined generated data and real data.The experimental results show that this method can effectively improve the prediction accuracy of seizures. The framework diagram is given in Fig. 1.It consists of five parts: Epilepsy brain electrical signal acquisition includes

preictal and interictal EEG data;The spectrum data from FFT; Features from DCGAN and classification using ELM.

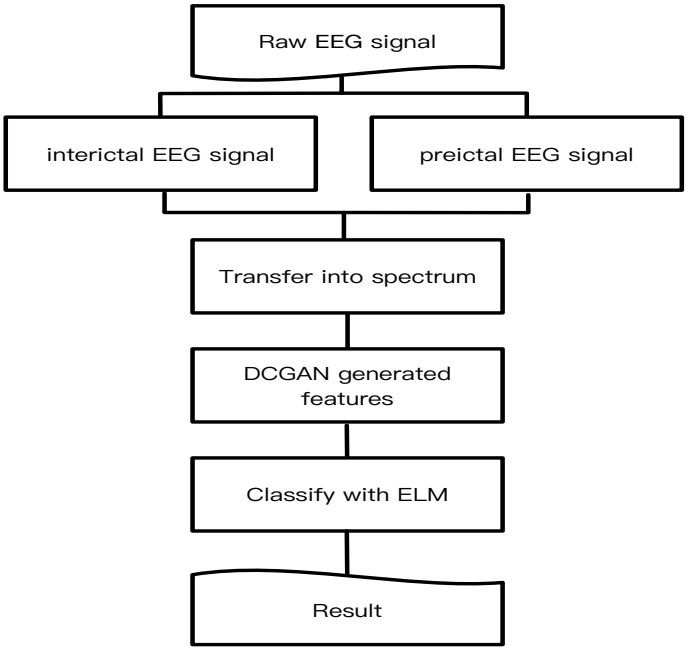

Fig. 1 Framework of epileptic seizure prediction method using generative features

From the point of feature expression, deep learning has excellent ability in image feature learning and extraction. DCGAN combines CNN with GAN, and combines the advantages of convolution neural network and anti neural network to learn multidimensional high level vector features from input data. DCGAN network has very high application value in the field of image generation. Therefore, when we do epileptic seizure prediction, we transform the EEG mode to maximize the ability of using the neural network. As an unsupervised learning method, DCGAN has more autonomy and depth in the feature extraction of epileptic brain signals, and the characteristics of epileptic data in higher dimensions are extracted from the training of its own structure. ELM is a unique classification method in the field of deep learning. It not only has the high performance of the convolution neural network for data classification, but also has the great learning and computing speed . As a single hidden layer feedforward neural network, ELM abandons the training mode of the traditional neural network, and the weights and biases between the input layer and the hidden layer are generated randomly. It only needs to calculate the weight and bias of the hidden layer to the output layer. To a great extent, the learning ability and learning speed of the network have been improved. At the same time, the generalized inverse theory is introduced, and the result can be obtained by a single calcu-

lation through the least square solution method without iteration. Therefore, after DCGAN feature extraction and generation, ELM is used as a classifier to classify epileptic EEG data.
Finally,we can get the classification accuracy of different periods of EEG data which were introduced as prictal and interictal stage. The result can show the great performence of our proposed method applied on this epilepsy prediction mission.

Table 2 The experiment precess Algorithm Framework description

Algorithm Framework
INPUT
Raw epilepsy EEG data, corresponding Label.
OUTPUT
Classification accuracy
BEGIN
for data **in** { Patient_1,Patient_2}
let
NFFT ; (The number of data points used in each block for the FFT)
overlap ; (The number of points of overlap between blocks)
Fs ; (The sampling frequency)
window; (A function or a vector of length NFFT, the default is window$_h anning()$.)
ComputationalFormula :
$G_f(\beta, u) = \int_R f(t)g(t-u)e^{-j\beta t}dt$
for number **in** training steps
Sample n training images, X
Compute n generated images, G
Compute discriminator probabilities for X and G
Label training images 1 and generated images 0
Cost = (1/n)sum[log(D(X)) + log(1 - D(G))]
Update discriminator weights, hold generator weights constant
Label generated images 1
Cost = (1/n)sum[log(D(G))]
Update generator weights, hold discriminator weights constant
Classify with Extreme Machine Learning.
Get the results.
END

2 Result

2.1 Dataset

This article is being used on data created by the University of Pennsylvania and Mayo Clinic Intracranial EEG data of two groups of epileptic patients with epilepsy collected from intracranial electroencephalogram. The intracranial electroencephalogram is common way used to determine the stage of epilepsy to prevent future seizures.

The collection of data using different numbers of electrodes, the sampling frequency of 5000Hz, the reference electrode is the extra-cranial electrodes.The data is divided into a period of ten minutes, the data categories are divided into preictal stage of seizure and interictal stage of seizure. Data Sections were chronologically numbered and the test data were randomly assigned. The pre-seizure consisted of data for one hour before the onset (1: 05-0: 05 pre-seizure). Data taken five minutes prior to seizure (1) Ensure adequate operation Time to patients, family members, medical staff to respond, promptly issued a warning and medication treatment; (2) to ensure that any guarantee data integrity to prevent the pre-attack warning information is missing the same episode interval data is divided into ten minutes for a total of one hour.Duration of the interictal data from the complete data records randomly selected and effectively avoid the pre-seizure and post-seizure signal effects in order to avoid epilepsy Of the pre-, interphase, exacerbation and post-exacerbation of the data interfered with, requiring data collection before and after the week can not have other seizures.

Human subjects were collected during the monitoring process is shorter than the time of the dog subjects were collected, the entire monitoring The procedure should be less than one week and the interphase data collection time should be strictly controlled at least four hours before or after the epileptic seizure.The interval data should be randomly selected from the canine subjects and the human subjects when the conditions are satisfied.

Table 2 The number of used data clips of each subject

Subject	$Patient_1$	$Patient_2$	channels	sample frequency
Train interictal	50	42	24	5000 Hz
Train preictal	18	18	24	5000 Hz
Test data	195	150	24	5000 Hz

2.2 Evaluation index

The evaluation index used in this experiment is shown in the following table. Including sensitivity, specificity, accuracy, precision, AUC etc. In the table, TP represents the number of positive samples correctly classified, TN represents the number of negative samples correctly classified, FP represents the number of

samples with negative classification errors, FN represents Positive examples are misclassified samples. If there are M positive samples and N negative samples, the classifier will get a score of samples judged to be positive after prediction, sort the scores from big to small, In this ordered sequence, the positive score is greater than the negative score and the AUC score is obtained.

Table 1 Table of evaluating indicators

Evaluation index	Defination	Function Definition
Sensitivity	Positive sample classified correctly	$Sensitivity = \frac{TP}{TP+FN} \times 100\%$
Specificity	Negative samples classified correctly	$Specificity = \ \times 100\%$
Precision	Correctly classified proportion	$Precision = \frac{TP}{TP+FP} \times 100\%$
Accuracy	All samples classified correctly	$Accuracy = \frac{TP+TN}{TP+FN+TN+FP} \times 100\%$
AUC	Area under the curve	$AUC = \frac{\sum_{ins_i \in positiveclass} rank_{ins_i} - \frac{M \times (M+1)}{2}}{M \times N}$

2.3 Experiment Result

In this experiment we use the training and testing dataset ratio of 3:1. Using FFT change the labeled data into spectrum patterns. All spectrum will be stratified sampling to obtain multiple images, wherein each picture set contains 6000 pictures, the proportion of generated positive samples and negative samples remain the same ratio, after dividing the training set and test set for each group of pictures. The training set data according to the true annotation is divided into two image sets were put into DCGAN for training, generating 6400 pictures of each class, the size normalized images generated for $28 \times 28$and the original training samples as ELM classifier training samples, and then to predict the feature map of the test set, the result of prediction the average values are shown in the following table.

Table 2 The classification results of patient1 with DCGAN method

Subjects	Sensitivity	Specificity	Precision	Accuracy	AUC
Rchannel	84.10%	97.02%	90.86%	93.66%	0.96
Gchannel	74.34%	94.69%	83.12%	89.40%	0.94
Bchannel	61.54%	94.94%	81.06%	86.25%	0.90
grey pic	70.52%	93.39%	78.97%	87.44%	0.91

It can be seen from Table 2 that the color spectrum generated by patient1 has the best classification effect on the R channel, and all the indexes are higher than 2.00%compared with the G channel with the second best performance. In particular, Precision The index is higher than the 7.74 %, which reflects the R channel of the R channel with different categories of color maps has the largest distinguishability among the three channels. As can also be seen from FIG. 7, the ROC curve obtained by processing the R channel The largest area under the same time, the added grayscale contrast experimental results, reflecting the direct use of grayscale classification results worse than the separate use of R channel and G channel classification results, which also shows that different types of grayscale in the distinction Degree has decreased.

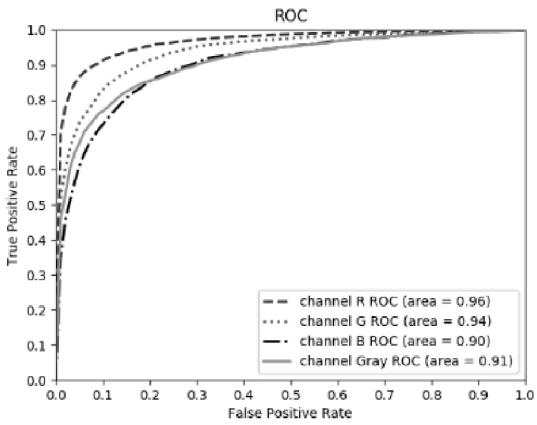

Fig. 7 ROC curve of patient1 feature of DCGAN method

Table 3 The classification results of patient2 with DCGAN method

Sbujects	Sensitivity	Specificity	Precision	Accuracy	AUC
Rchannel	91.34%	97.34%	93.83%	95.50%	0.98
Gchannel	81.21%	93.47%	84.62%	89.71%	0.94
Bchannel	88.05%	95.97%	90.63%	93.54%	0.97
Grey pic	75.68%	91.19%	79.17%	86.43%	0.92

As with patient1, patient2 generated the best classification on the R channel, at least % above the B channel, and 3.83% above the Precision index, and at the same time the grayscale performance Still not as good as the R channel classification performance. In order to verify the validity of the experiment, a comparative experiment [34] based on coherence average (CM), spectral entropy (SE), spectral correlation power (SRP) and three feature fusion was added. The classification results obtained are shown in the following table.

Table 4 The classification results of patient1 with DCGAN method

Subjects	Sensitivity	Specificity	Precision	Accuracy	AUC
CM	71.61%	57.34%	37.13%	61.05%	0.66
SE	80.66%	52.71%	37.50%	59.98%	0.69
SRP	75.97%	63.43%	42.22%	66.69%	0.74
(CM,SE,SRP)	84.99%	72.39%	51.84%	75.54%	0.81

The table above reflects the classification results of CM, SE, SRP and patient-specific feature fusion of patient1. It can be seen that the classification result obtained by using the single feature is poor with the maximum Sensitivity of 80.66%, which is derived from the classification result of SE feature The convergence of the 3 features led to a significant increase in classification performance, as can be seen in Figure 9. For Sensitivity, the value increased by 4.3% after feature fusion, and Accuracy also gained at least 8.85% However, there is still a certain gap between the two methods as a whole.

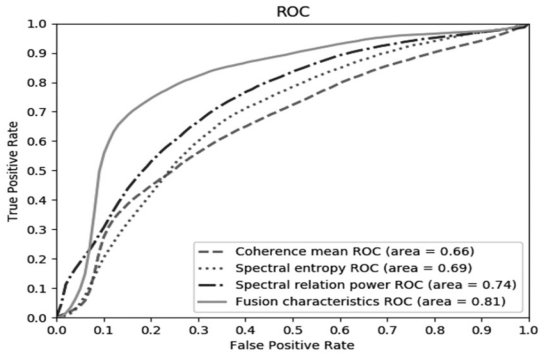

Fig. 9 ROC curve of patient 1 with CM, SE, SRP features

Table 5 The classification results of patient2 with DCGAN method

Subjects	Sensitivity	Specificity	Precision	Accuracy	AUC
CM	56.11%	45.71%	31.39%	48.90%	0.52
SE	69.38%	47.61%	36.96%	54.29%	0.62
SRP	62.66%	46.96%	34.34%	51.78%	0.59
(CM,SE,SRP)	72.58%	54.46%	41.47%	60.16%	0.71

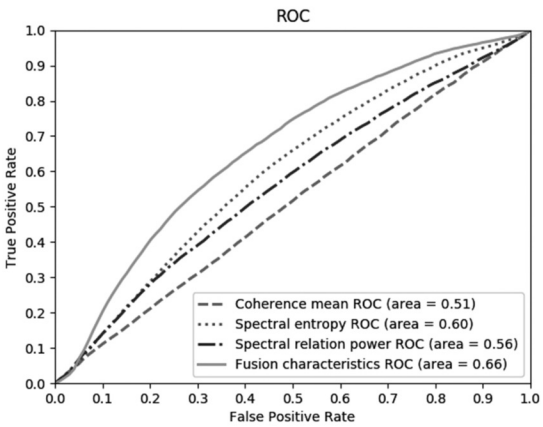

Fig. 10 ROC curve of patient 2 with CMSESRP features

For patient2, the three extracted features do not reflect the differences between samples well. For the single-feature classification, the SE feature obtained

the highest Sensitivity, 6.72% higher than the SRP, but the Accuracy Only 54.29%, while the fusion feature achieved a classification result of at least 3.20% improved on Sensitivity and an increase of 5.87% on Accuracy. As can be seen from Figure 10, the area under the ROC curve obtained for classification of the fusion features is the largest, which also reflects the improvement in classification performance over the single feature.

3 Conclusion

The research on the prediction of seizures helps to broaden the cognition of brain science in human and also has important physiological significance for patients with epilepsy.However, the study of epilepsy is facing the problem of poor accuracy of prediction of seizures, inadequate data and diseases? Personal information security and other issues.In this paper, the deep convolutional generative adversarial networks (DCGAN) are used to extract and generate the features of the original EEG features, and finally use the extended learning machine (Extreme learning machine, ELM) to classify the different stages of seizures, as early as possible at an earlier warning point to help patients and their families to timely detection and avoid seizures. For medical data will always encounter inadequate experimental data and patients The information desensitization problem, using DCGAN model generation, can solve these two problems well.The method proposed in this paper can predict the epileptic seizures accurately and play an enlightening role in the study of other diseases such as depression, Parkinson's disease and ADHD.

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Appendix